

2. Stochastic control problem

As already mentioned in the introduction, the turnpike phenomenon is a significant principle in optimal control theory. In fact, it establishes that during a specific interior subinterval of a given time horizon, the dynamic control strategy can be approximated with arbitrary precision by the solution of a (generally less complex) static optimization problem. This insight greatly reduces the computational effort typically associated with dynamic control strategies.

In our framework, we start by considering the continuous trajectories of the interacting agents. We denote by $x_k(t) \in \mathbb{R}^d$ the position of agent k at time t , with $k = 1, \dots, N$, and the time interval is defined as $[t_0, T]$. The dynamics of the agents are described by the following stochastic differential equation (SDE)

$$dx_k(t) = \left(\frac{1}{N} \sum_{\ell=1}^N p(x_k(t), x_\ell(t)) (x_\ell(t) - x_k(t)) + u_k(t) \right) dt + \sigma dW_k(t), \quad (1)$$

where the term $\sigma dW_k(t)$ represents the stochastic noise component, with $W_k(t)$ being standard d -dimensional independent Brownian motions. The initial condition for each agent k is given by $x_k(t_0) = x_k^0 \in \mathbb{R}^d$, where x_k^0 is a stochastic variable, accounting for randomness in the initial state distribution. The interaction between agents is mediated by a nonlinear interaction kernel $p(x_k(t), x_\ell(t))$ which describes the forces governing their behavior. For further examples and theoretical background, we refer to [7]. The system operates within a well-defined probabilistic framework. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space, where Ω represents the sample space, $\mathcal{F}$ is the σ -algebra of measurable events, and $\mathbb{P}$ is the probability measure. The stochastic processes $W_k(t)$, are adapted to the filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq t_0}$, which represents the natural filtration generated by the Brownian motions. This filtration captures all the information available up to time t , ensuring that the evolution of the stochastic processes depends on the past without anticipating future information.

The dynamics of the agents are thus driven by both deterministic controls $u_k(t)$ and the stochastic influences represented by the Brownian motions $W_k(t)$. The control $u_k(t)$ is applied individually to each agent, with the aim of minimizing a cost functional over a finite time horizon. The control objective is to drive the system towards a desired state $\bar{x} \in \mathbb{R}^d$ while minimizing the associated cost. The latter (to be minimized with respect to the control term) incorporates a tracking term, which measures the deviation of each agent from the desired state $\bar{x}$, and a regularization term to penalize the magnitude of the control. Specifically, the cost functional, which depends on $\mathbf{x} = (x_1, \dots, x_N)$, $\mathbf{u} = (u_1, \dots, u_N)$, is given by

$$\begin{aligned} \mathcal{C}(\mathbf{x}, \mathbf{u}) &= \int_{\Omega} \int_{t_0}^T \frac{1}{N} \sum_{k=1}^N (\|x_k(t, \omega) - \bar{x}\|^2 + \gamma \|u_k(t)\|^2) dt d\mathbb{P}(\omega) \\ &= \mathbb{E} \left[\frac{1}{N} \int_{t_0}^T \sum_{k=1}^N (\|x_k(t) - \bar{x}\|^2 + \gamma \|u_k(t)\|^2) dt \right]. \end{aligned} \quad (2)$$